

A Dynamic Control System for Server Processor Direct Liquid Cooling

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Abstract— Direct liquid cooling enables the development of IC, especially processors in servers, towards higher density, which demands more efficient means of heat removal. To exploit the capability, control system that dynamically balances between cooling power and energy consumption is critical for the operational safety and cost. This paper presents the design, implementation and characteristic of a dynamic control system for the server processor direct liquid cooling system designed for high density data center. Structure of the control system has been designed. Models for the liquid cooling components have been established and identified. Control algorithm based on predictive model and deep neural network has been implemented. Experimental validation has been conducted. Evaluation criteria has been established to characterize the overall system performance for benchmarking. The results validated the effectiveness of the proposed dynamic control system, and showed the advantages of the designed deep explicit MPC in terms of cooling effectiveness and energy efficiency. Primary loop mean PUE reaches 1.0023 with -0.96°C mean overheat level. It has been reduced over 1.2°C overheat level and 8 times of overheat time percentage than the traditional open loop control under similar energy budget. Energy consumption has been reduced by 21.7% than close loop proportional control under similar average overheat level.

Index Terms— direct liquid cooling, dynamic control, model predictive control, neural network, data center,

I. INTRODUCTION

Modern society relies heavily on the information technology. The recent progress in 5G wireless communication, machine learning and computational science requires more powerful and denser data center servers [1-4]. The demand drives the processor manufacturers, e.g. Intel, towards extreme transistor integration, clock frequency and multiple cores design. If we look at the trend of microprocessor data over the past 42 years[5], it can be seen that Moor's law is facing significant "thermal crisis"[6] that "the thermal design power is now limiting the growth in computing performance"[7]. To release more power, cooling solution has begun shifting from traditional widely used air cooling technology [8-10] to novel liquid cooling technology [11-14]. Meanwhile, the major concern of the energy consumption on cooling facilities [15, 16] has started being relieved. Compact micro-fluid direct liquid cooling [17] devices made of Cu [18-20] or packages made of Si[21, 22] has been manufactured and

studied. Compatible mini heat exchanger has been designed and characterized [23]. Significant reduction of thermal resistance and improvement of energy efficiency has been demonstrated.

Despite progresses on the cooling devices and systems, it has been reported [24, 25] as a technical challenge to overcome that implementing dynamic control on the cooling system to handle varying workloads, power dissipation surges and fast transients. Either air cooling [26, 27] or liquid cooling [28, 29] in data center have shown improvements in temperature profile and energy efficiency in the past studies on dynamic control. Traditional cooling system widely adopted the proportional-integral-derivative (PID) controller [30, 31], which reactively changes manipulated variable according to feedback variable. Technical challenges lie in the complex thermal dynamics in the system such as nonlinearity, inter-loop coupling and fast changing disturbances. Active disturbance rejection control algorithm for air-cooled 1U server has been proposed to reduce up to 45% of energy comparing to PID controller [32].

Comprehensive review [33] reported that the model predictive control (MPC) has shown the potential of improvement in transient response, robustness to disturbances and energy saving in the application of heating, ventilation and air conditioning system which shares certain similarities with the data center cooling system. One of the main drawbacks of MPC that prevents its application is "the need to solve a mathematical (optimization) program on-line to compute the control action", which is addressed by explicit MPC [34]. Explicit MPC allows solving optimization problem off-line within given boundary, at a cost of storage space. Deep neural network is a good candidate to approximate the explicit form of solution such that neural computing accelerator can infer the control action efficiently in a dynamic environment.

This paper presents the approach to establish an explicit MPC controller for liquid cooling system based on deep neural network, including modeling for system dynamics, structure design of the system, and implementation of the controller. Experimental results demonstrate the advantages of the designed deep explicit MPC in terms of key evaluations in cooling effectiveness and energy efficiency over PID controllers. Challenges have been addressed in this paper that simulation modeling using MATLAB/Simulink and Simscape toolbox, system integration among NI LabVIEW, python and MATLAB, and obtaining the deep neural network representation for explicit MPC controller.

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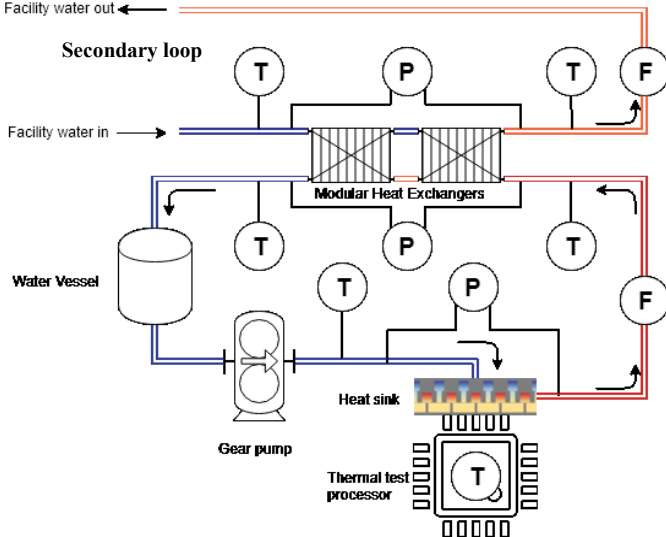


Fig. 1. System diagram of the experimental platform

II. DESIGN OF LIQUID COOLING SYSTEM

Target for control is a liquid cooling system for high power density electronic devices in data center. As shown in Fig. 1, this system consists of primary loop and secondary loop in a cascaded architecture. The working fluid in the primary loop is 10 μ m-filtered de-ion water, and the working fluid in the secondary loop is facility water. Tap water is used as working fluid for the secondary loop without further process instead of recycled facility water due to limited facility water access. Devices in this experimental platform are listed: a jet impingement liquid cooling packaging heat sink as the key control element, thermal test processor, heat exchanger, pump, water reservoir and necessary sensors to monitor the pressures, flow rates and temperatures of many points in the entire system.

The target element of control is the cooling package for several high-power-density logic processors. This heat sink is a TCB-stack of three micro-etched Silicon dies enclosed in an aluminum casing. The cavity forms multiple micro-jet impingement slots surrounded by draining trenches such that crossflow passing through the jet area is avoided to improve the cooling efficiency. The overall hydraulic pressure loss can be reduced since the paralleled flow paths are shortened and the total flow rate can be reduced thanks to higher heat absorption

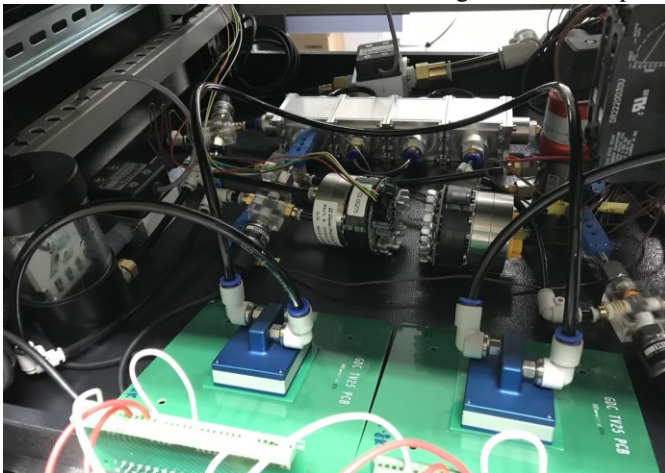


Fig. 2. Si-based liquid cooling package prototype

TABLE I
SPECIFICATIONS OF SENSORS AND DATA ACQUISITION SYSTEM

Item	Measure	Unit	Range	Accuracy	Rate (Hz)
NI PXIe-6355	DC voltage	V	[-10, 10]	607 μ	2k
McMillan 106-5D	flow rate	L/min	[0.05, 0.5]	0.005	
McMillan 106-8D	flow rate	L/min	[0.5, 5]	0.05	
Sorensen DLM 600-5E	DC voltage	V	[0, 600]	3.1	
	DC current	A	[0, 5]	0.05	
NI PXIe-6239	DC current	mA	[-20, 20]	15.8E-6	2k
Huba 692	pressure	kPa	[0, 50]	0.625	
Keyence AP-12S	pressure	kPa	[0, 100]	0.5	
NI PXIe-4353	temperature	$^{\circ}$ C	[0-200]	0.28	4
Watlow type-T	temperature	$^{\circ}$ C	[0-200]	1.0	
NI PXIe-4339	quarter bridge	-	[-1, 1]	0.1%	2k
NI PXIe-6739	DC voltage output	V	[-10, 10]	2.94 μ	2

efficiency of the jet impingement, compared with straight micro-channel heat sink. The sample is designed, manufactured and tested in previous work [35], as shown Fig. 2 bottom side. Fully developed jet impingement is formed in the package to create strong heat convection. Flow path and crossflow are optimized by the proper design of the structure.

A thermal test chip without logic function is packaged inside for testing purposes. A resistive heater simulates the activities of the logic chip and a Pt RTD sensor detects the internal temperature. Thick thermal interface materials are used to prevent cracking Si dies, which introduces additional thermal resistance to the system.

Customized modular heat exchanger used in the experiment is shown in Fig. 2 upper side, which is designed and characterized in another work [36]. The design flow rates of this modular heat exchanger are 1.25L/s for the primary loop side and 2.5L/s for the secondary loop side at nominal 500W power, with pressure drop of 16kPa and 11kPa respectively. The overall effectiveness coefficient is 0.9. For transferring 230W heat, the pumping power to overcome the pressure loss is only 0.075W.

Besides of the home-made cooling package and modular heat exchanger, other components and equipment in the system are commercially available. The main components and their key parameters are described as follows. Accuracies and sampling rates of the sensors are listed in TABLE I.

A rack-mounted Sorensen DLM600-5E power supply with 0-600V supply voltage and 0-5A current is used. The voltage set point is adjusted by a 0-5VDC analog signal, and real-time feedback of voltage and current measurement is transmitted by 0-5VDC analog voltage signal. An external relay is used to protect the circuit.

A DDP Silencer 1000 gear pump is adopted to pump water in the primary loop. The power supply is 24V DC power, which is provided by NI PXI-4110 module so that real-time pumping power is monitored. The pump speed is adjustable by a 0-5VDC control signal with pulse feedbacks signal in a rate of 4 pulses per revolution.

An NI PXIe-8840 controller operated in Windows 10 Pro is hosting the implementation of the online stage of the control

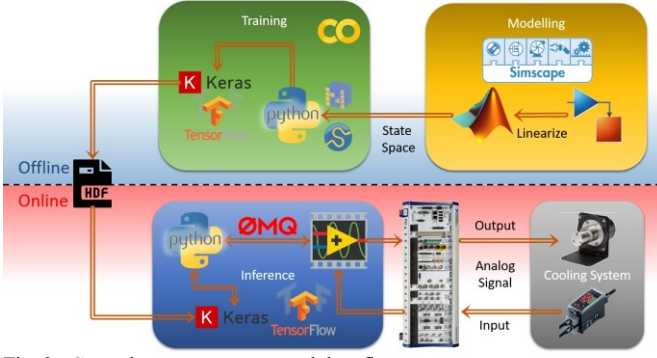


Fig. 3. Control system structure and data flow

system. Fig. 3 illustrates the structure and data flow of the control system including online stage and offline stage. Because of the digital computer-based controller, this control system runs in discrete-time way at an operating cycle frequency of 10Hz. Sampling, computing and acting are executed sequentially and repeatedly. First, each NI module measures sensor signals in independent sampling rate, and the LabVIEW main loop aligns the data by applying filters and/or first-order holders. Meanwhile, a python server that infers the control action via deep neural network is standing by and sets up a request-reply message queue with LabVIEW. When a request containing the measurements is raised to the server, python formats the received message into data array, and the neural network model, whose structure and weights are defined by a stored HDF5 file, infers the control action from the data array. At last, Python replies LabVIEW with a message containing the control action, and the LabVIEW performs control physically.

The neural network model is defined and trained in offline stage as shown in Fig. 3 upper side. The model is based on TensorFlow backend and built by Keras API, and the training is conducted on cloud-based computational platform Colab. Constructing computational graph for training the proposed explicit model predictive controller, which will be described in Section IV, requires modeling of the real system. Here, the model is first built in MATLAB/Simulink environment with the help of Simscape toolbox, and then linearized to state space representation automatically. The modeling details will be described in Section III.

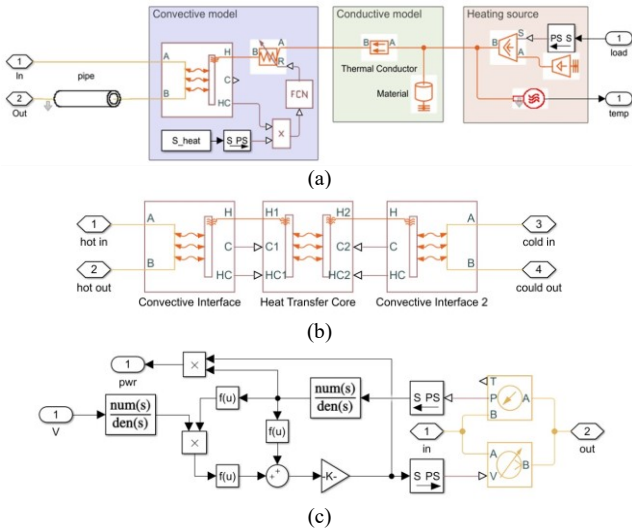


Fig. 4. Simulation model for liquid cooling system components in MATLAB. (a) cooling package. (b) heat exchanger (c) pump.

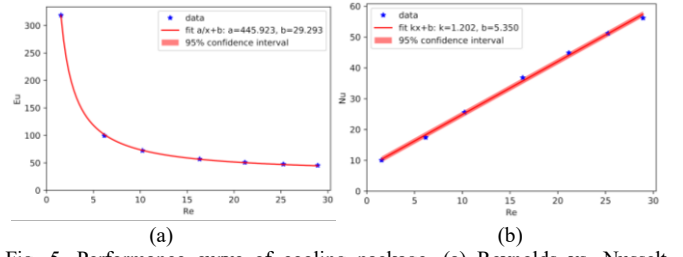


Fig. 5. Performance curve of cooling package. (a) Reynolds vs. Nusselt number. (b) Euler vs. Nusselt number

III. SYSTEM MODELING

The modeling of liquid cooling system starts with modeling of components, including pump, cooling package and heat exchanger implemented in MATLAB/Simulink by using Simscape, which is shown in Fig. 4.

Three components has been build to form the model for the cooling package in Fig. 4(a), namely the convection module, conduction module and source module. For convection module, Euler number in terms of Reynolds number is used to estimate the pressure difference across the component, and Nusselt number is used to characterize the dynamics of heat transfer:

$$\Delta p = \text{Eu}(\text{Re}) \frac{\dot{m}^2}{2\rho S_{\min}^2} \quad (1)$$

$$C_p \rho V \frac{dT_{\text{out}}}{dt} + \dot{m}(T_{\text{out}} - T_{\text{in}}) = \dot{Q} = \frac{\text{Nu} k S_h}{D_h} (T - T_{\text{in}}) \quad (2)$$

A biased inverse proportionality is founded by data-driven model between Euler number and Reynolds number for the cooling package in this paper, shown in Fig. 5(a). And similarly, a linear relationship is founded between Nusselt and Reynolds number as shown Fig. 5(b). A narrow 95% confidence interval indicates the goodness of fit.

Bulk parameter model is used to describe the temperature dynamics at junction and surface and heat conduction:

$$\dot{Q} = k \frac{S_h}{t} (T_j - T) = P_{\text{load}} - C \rho S_h t \frac{dT_j}{dt} \quad (3)$$

A variable ideal heat source with controllable heat flux to the surface is used in source module. The heat flux is linear related to the CPU clock usage.

An e-NTU model with an effective parameter ε is adopted for the heat exchanger in Fig. 4(b):

$$Q_1 = -Q_2 = \varepsilon Q_{\max}, 0 < \varepsilon < 1 \quad (4)$$

where the $Q_{\max}$ (highest potential heat transfer) is proportional to the temperature difference between two fluids $Q_{\max} = C_{\min}(T_{1,\text{in}} - T_{2,\text{in}})$, and an imperial formula for shell and tube heat exchanger is adopted to estimate ε :

$$\varepsilon = \frac{2}{1 + C_{\text{rel}} + \sqrt{1 + C_{\text{rel}}^2} \frac{1 + \exp(-\text{NTU} \sqrt{1 + C_{\text{rel}}^2})}{1 - \exp(-\text{NTU} \sqrt{1 + C_{\text{rel}}^2})}} \quad (5)$$

where here uses overall thermal resistance between the two fluids to estimate the number of transfer unit (NTU):

$$\text{NTU} = \frac{U_{\text{overall}} A_{\text{heat}}}{C_{\min}} = \frac{1}{C_{\min} R_{\text{overall}}} \quad (6)$$

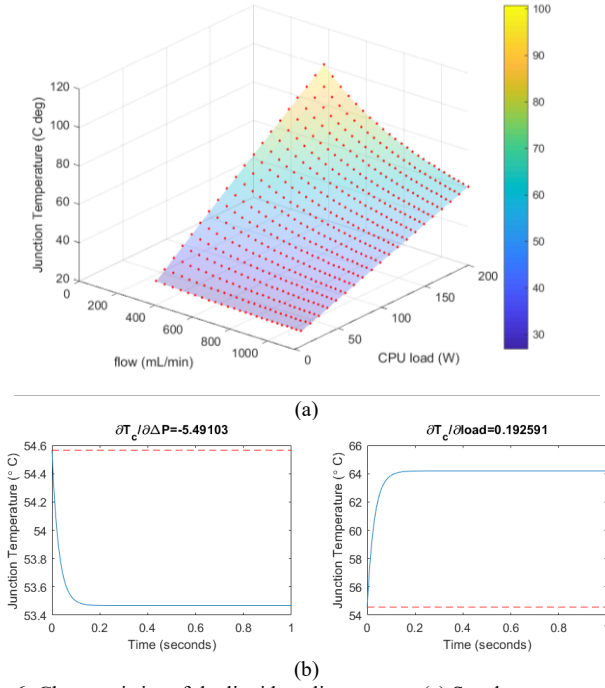


Fig. 6. Characteristics of the liquid cooling system. (a) Steady-state response surface. (b) Transient response of pressure step test and load step test.

The flow rate in terms of pressure head and pump revolution rate is empirically described by the model in Fig. 4(c):

$$Q = f(P, V) = d \left(\frac{aP+c}{d} V + \frac{bP}{d} + 1 - \exp\left(\frac{aP+c}{d} V\right) \right) \quad (7)$$

where $a = 101.4$, $b = -918.7$, $c = 1144$, and $d = -702.9$.

System model synthesis is implemented with the help of governing equations for energy conservation and mass conservation, which is implemented by Simscape toolbox. Incompressibility is assumed for liquid coolant since the local sonic speed (propagation of motion and vibration) is much faster than the heat transfer dynamics. Well enclosed and well insulated assumptions are made for the pipe networks, so there is no leakage and external heat transfer process in the model.

Based on the established model, the temperature response surface (Fig. 6(a)) as well as step response dynamics are visualized (Fig. 6(b)). By subjecting the tabulated nonlinear steady-state response $\mathbf{y}_{ss}$ from the model, one can obtain linearized transient dynamics from the established MATLAB model. A nonlinear state-space representation model which hybrid the nonlinear steady-state response with linearized transient response is established:

$$\begin{aligned} \tilde{\mathbf{x}}[k+1] &= \mathbf{A} \tilde{\mathbf{x}}[k] + \mathbf{B} \mathbf{u}[k+1] + \mathbf{H} \mathbf{d}[k+1] \\ \mathbf{y}[k] &= \mathbf{C} \tilde{\mathbf{x}}[k] + \mathbf{y}_{ss}(\mathbf{u}[k], \mathbf{d}[k]) \end{aligned} \quad (8)$$

the manipulated variable $\mathbf{u}$ contains the pump signal V in (7), disturbance variable $\mathbf{d}$ contains the power load of P_{load} in (3), the control variable $\mathbf{y}$ contains the core junction temperature T_j in (3), and the state variable $\tilde{\mathbf{x}}$ has a space of $\mathbb{R}^4$.

IV. DEEP EXPLICIT MODEL PREDICTIVE CONTROL

The proposed deep explicit model predictive control is based on a more general assumption of discrete time nonlinear model of controlled plant with measurable disturbance:

$$\begin{aligned} \mathbf{x}[k] &= f(\mathbf{x}[k-1], \mathbf{u}[k], \mathbf{d}[k]) \\ \mathbf{y}[k] &= h(\mathbf{x}[k]) \end{aligned} \quad (9)$$

$$\begin{aligned} \text{subject to: } g(\mathbf{x}[k], \mathbf{u}[k], \mathbf{d}[k]) &\leq 0 \\ \bar{g}(\mathbf{x}[k], \mathbf{u}[k], \mathbf{d}[k]) &= 0 \end{aligned}$$

where the $\mathbf{x} \in \mathbb{S} \subseteq \mathbb{R}^s$ is the state variable with s dimensions, the $\mathbf{u} \in \mathbb{M} \subseteq \mathbb{R}^m$ is the manipulated variable with m dimensions, the $\mathbf{d} \in \mathbb{D} \subseteq \mathbb{R}^d$ is the disturbance variable with d dimensions, and $\mathbf{y} \in \mathbb{O} \subseteq \mathbb{R}^o$ is process variable with o dimensions. The forward step function $f: \mathbb{S} \times \mathbb{M} \times \mathbb{D} \rightarrow \mathbb{S}$ and the output function $h: \mathbb{S} \rightarrow \mathbb{O}$ are given. The operation $\circ[k]$ represents temporal indexing of any variable at time k . The hybrid state-space representation in Section III can be easily converted to this paradigm by state augmentation.

Then the n -step forward prediction is defined recursively:

$$f_i(\mathbf{x}_0, \mathbf{u}[1:i], \mathbf{d}[1:i]) = \begin{cases} f(f_{i-1}, \mathbf{u}[i], \mathbf{d}[i]), & i > 0 \\ \mathbf{x}_0, & i = 0 \end{cases} \quad (11)$$

where $f_i: \mathbb{S} \times \mathbb{M}^n \times \mathbb{D}^n \rightarrow \mathbb{S}$ is forward function, and $[\circ:]$ is temporal indexing operator $\mathbf{v}[1:i]^T = [\mathbf{v}[1]^T \ \dots \ \mathbf{v}[i]^T]$.

The disturbance variable $\mathbf{d}$ is assumed to follow Markov process with given probability $P(\mathbf{d}[i+1]|\mathbf{d}[i])$. In this definition, the joint probability of future n -step disturbances in term of initial disturbance $\mathbf{d}_0 = \mathbf{d}[0]$ is given by:

$$\rho_{\mathbf{d}_0} = P(\mathbf{d}[1:n]|\mathbf{d}[0]) = \prod_{i=1}^n P(\mathbf{d}[i]|\mathbf{d}[i-1]) \quad (12)$$

Reference trajectory for tracking is given as $\mathbf{r}[1:n] \in \mathbb{O}^n$ which has same dimension as $\mathbf{y}$. Define $\mathbf{v} = [\mathbf{x}_0^T, \mathbf{r}[1:n]^T, \mathbf{d}_0^T]^T$ in a polytopic set $\mathbb{V} = \mathbb{S} \times \mathbb{O}^n \times \mathbb{D}$. An optimization program can be set up as (13) to plan the future n -step manipulated variables. Regularization term on control variable with a linear transformation coefficient $\mathbf{E}$ is added.

The controller iteratively applies $\mathbf{u}^*[1]$ from the computed $\mathbf{u}^*[1:n]$ that minimizes J at current $\mathbf{v}$, which is computational costly for nonlinear systems due to the nature of optimization-in-a-loop. An alternative solution is off-line computing and storing the optimal $\mathbf{u}^*[1]$ within a given range of $\mathbf{v}$ to reduce online computational cost. The results are usually tabulated or formulated as a collection of piecewise linear functions:

$$\begin{aligned} J(\mathbf{v}, \mathbf{u}[1:n], \mathbf{d}[1:n]) &= \|\mathbf{r}[1:n] - \mathbf{h}_{1:n} \circ \mathbf{f}_{1:n}(\mathbf{x}_0, \mathbf{u}[1:n], \mathbf{d}[1:n])\| + \|\mathbf{E} \mathbf{u}[1:n]\| \\ \min_{\mathbf{u}[1:n]} \mathbb{E}_{\mathbf{d}[1:n] \sim \rho_{\mathbf{d}_0}} J(\mathbf{v}, \mathbf{u}[1:n]) &= \int_{\mathbb{D}^n} \rho_{\mathbf{d}_0} J(\mathbf{v}, \mathbf{u}[1:n], \mathbf{d}[1:n]) d\mathbf{d}[1:n] \end{aligned} \quad (13)$$

subject to:

$$\begin{aligned} \mathbf{g}_{1:n}(\mathbf{f}_{1:n}, \mathbf{u}[1:n], \mathbf{d}[1:n]) &\leq 0, \forall \mathbf{d}[1:n] \sim \rho_{\mathbf{d}_0} \\ \bar{\mathbf{g}}_{1:n}(\mathbf{f}_{1:n}, \mathbf{u}[1:n], \mathbf{d}[1:n]) &= 0, \forall \mathbf{d}[1:n] \sim \rho_{\mathbf{d}_0} \end{aligned}$$

where $\mathbf{f}_{1:n}: \mathbb{S} \times \mathbb{M}^n \times \mathbb{D}^n \rightarrow \mathbb{S}^n$ is the vector form of concatenation of $\{f_i|i \leq n, i \in \mathbb{Z}^+\}$, $\mathbf{h}_{1:n}: \mathbb{S}^n \rightarrow \mathbb{O}^n$ is vectorized form of h , and $\mathbf{g}_{1:n}, \bar{\mathbf{g}}_{1:n}: \mathbb{S}^n \times \mathbb{M}^n \times \mathbb{D}^n \rightarrow \mathbb{R}^n$ is vectorized form of $g, \bar{g}$.

$$\min_{\theta} \mathbb{E}_{\mathbf{v} \sim \mathcal{U}(\mathbb{V}), \mathbf{d}[1:n] \sim \rho_{\mathbf{d}_0}} [J + \lambda G] = \iint_{\mathbb{V} \times \mathbb{D}^n} \rho_{\mathbf{d}_0} [J(\mathbf{v}, \boldsymbol{\pi}_{\theta}(\mathbf{v}), \mathbf{d}[1:n]) + \lambda G(\mathbf{v}, \boldsymbol{\pi}_{\theta}(\mathbf{v}), \mathbf{d}[1:n])] d\mathbf{v} d\mathbf{d}[1:n] \quad (18)$$

$$\mathbf{u}^*[1] = \begin{cases} \mathbf{F}_1 \mathbf{v} + \mathbf{b}_1, & \mathbf{H}_1 \mathbf{v} + \mathbf{k}_1 \geq 0 \\ \vdots & \vdots \\ \mathbf{F}_N \mathbf{v} + \mathbf{b}_N, & \mathbf{H}_N \mathbf{v} + \mathbf{k}_N \geq 0 \end{cases} \quad (14)$$

Traditionally, explicit solution for (14) is obtained by multi-parametric program. However, for nonlinear system, the optimal solution is not easily characterized while suboptimal approximation is favorable in the engineering point of view. Assume an arbitrary function approximator $\boldsymbol{\pi}_{\theta}: \mathbb{V} \rightarrow \mathbb{M}^n$ parametrized by θ . The purpose is to let $\boldsymbol{\pi}_{\theta}$ approaches to the solution of the multi-parametric program by using a temporal independent θ within the given polytopic set $\mathbb{V}$. Such that:

$$\mathbf{u}^*[1:n] = \boldsymbol{\pi}_{\theta}(\mathbf{v}) \quad (15)$$

Recent great progress in the deep neural network shows that approximating large and complex nonlinear function with over millions of input dimensions are possible, as the development of computational capability, computing infrastructure, big data management and efficient structures of the deep neural network. The network structure is a stack of p fully connected residual layers. Each residual layer is defined as in (16). ReLU activation function is used to prevent gradient vanishing except the last layer which is sigmoidal.

$$\mathbf{T}_k = \mathcal{R}_{\mathbf{w}_k}(\mathbf{T}_{k-1}) = \text{ReLU}(\mathbf{w}_k \mathbf{T}_{k-1}) + \mathbf{T}_{k-1} \quad (16)$$

where $\mathbf{T}_k$ is the tensor of k -th layer, $\mathbf{w}_k$ is the tensor of weights, and $\text{ReLU}(\circ)$ is defined as $y = \max(0, x)$ elementwise. Parameter θ is the flattened concatenation of $\{\mathbf{w}_k | 1 < k < p\}$.

To adapt constrained optimization problem into standard deep learning framework, Lagrangian relaxation is used that adds the Lagrangian term G with a Lagrangian relaxation coefficient λ into cost function. The Lagrangian term punishes any violation of the constraints with a non-negative value which is defined as:

$$G((\mathbf{v}, \boldsymbol{\pi}_{\theta}(\mathbf{v}), \mathbf{d}[1:n])) = \|\text{ReLU}(\mathbf{g}_{1:n})\| + \|\bar{\mathbf{g}}_{1:n}\| \quad (17)$$

Finally the multi-parametric program is converted into a statistical optimization problem about θ as in (18), or “training” in the machine learning domain. The system diagram is shown

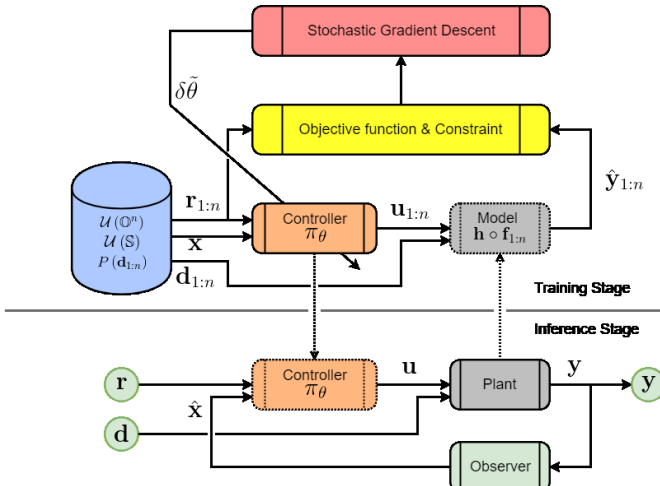


Fig. 7. Block diagram of deep explicit control algorithm.

in Fig. 7. Stochastic gradient descent (SGD) method with scheduled reducing learning rate, scheduled increasing Lagrangian relaxation λ is adopted. The training data are out-product of samples from $\mathcal{U}(\mathbb{V})$ and segments of Markov chain Monte Carlo (MCMC) based on $\rho_{\mathbf{d}_0}$. In the scope of this paper, the observer is derivable from the system model, which continuously samples $\mathbf{y}$ to estimate the state of the plant $\hat{\mathbf{x}}$. The model of the plant is given by (8). In the inference stage, the controller assumes future reference output maintains unchanged.

This control system is also compatible with observers and models that are identified during usage. An alternative point of view considers the physical model $\mathbf{h}_{1:n} \circ \mathbf{f}_{1:n}$ as modification of cost function, which could possibly form a generative adversarial network structure for co-training if represented by another neural network.

V. EXPERIMENTAL RESULTS

The testing range of the experiment is chosen based on:

- The pump signal $U_p \in [U_{\min}, U_{\max}]$ subjects to hardware limitation
- Processor load power $P_l \in [P_{\min}, P_{\max}]$ subjects to hardware limitation
- The core junction temperature $T_j \leq 65^\circ\text{C}, \forall U_p, P_l$ for long term safety, especially when $U_p = U_{\min}$ and $P_l = P_{\max}$.
- For controllability, reference set point T_r satisfies $T_j|_{U_p=U_{\max}, P_l=P_{\max}} \leq T_r \leq T_j|_{U_p=U_{\max}, P_l=P_{\max}}$

Despite being greatly narrowed down, the chosen working range reveals the features of the test platform and the dynamics of the control algorithm. The results from experiments in this working range can be reasonably extrapolated to certain wider working conditions given no phase change of the system. Finally, the working range is chosen as $U_p \in [0.3, 1.5]\text{V}, P_l \in [45, 50]\text{W}, T_r = 55^\circ\text{C}$.

At the beginning of the experiment, the system is initialized to the thermal equilibrium steady state under maximum pump speed. In each session of experiment, the same load schedule is applied for 700s as shown in Fig. 8, which is generated randomly. Different control strategies are applied based on this

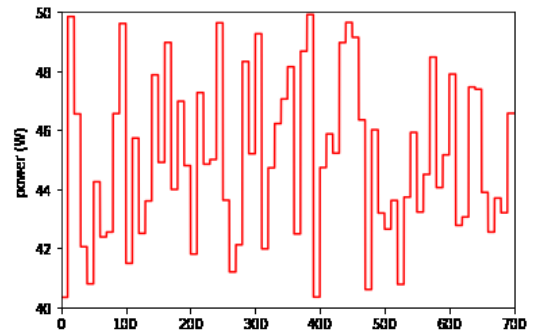


Fig. 8. Processor load schedule in experiment.

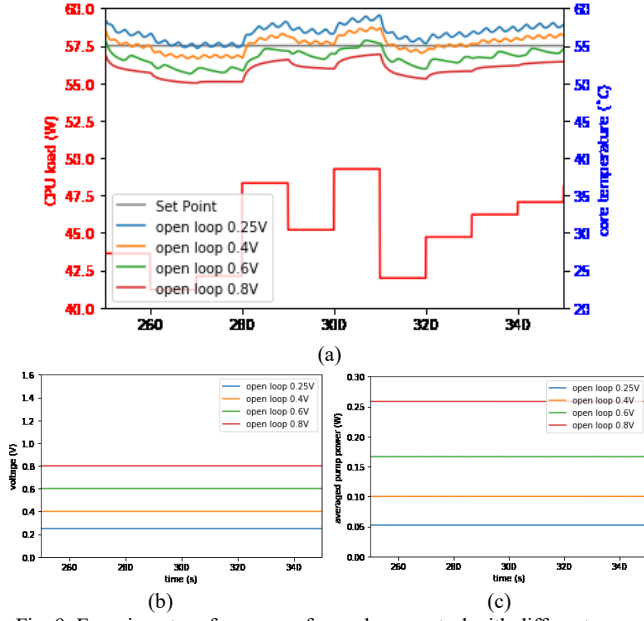


Fig. 9. Experiment performance of open loop control with different pump signal levels. (a) core junction temperature, (b) pump signal, (c) averaged pumping power.

load schedule, and the core temperature, pump signal and averaged pump power are measured correspondingly.

We first demonstrate the system performance with open loop control. This control strategy is simply setting constant pump signal regardless of temperature changes. The pump signals selected in the four experiments are 0.25V, 0.4V, 0.6V and 0.8V. Fig. 9 shows the details of the experiment between 250s and 350s as most of features are revealed. For low pump signal like 0.25V, the core temperature is almost always larger than the set point. For high pump signal like 0.8V, the core temperature is less than the set point. However, the average pumping power is almost 5 times comparing with the 0.5V pump signal case. Therefore, in dynamic load condition, it is hard to simultaneously keep cool core and save pumping energy without controller.

Next, we examine the performance of a traditional proportional controller in this system. The proportional controller gives pump signal in proportional to the difference between the real-time core temperature and the set point. In this controller, we applied a bias on the proportional controller such that the controller will always output full speed of pump when the core temperature is above the set point. The minimum pump signal voltage is 0.2V in order to prevent complete stop of flow. Fig. 10 shows the performance of the proportional controllers with different proportional gains K_p in the same time segment. It reveals that close loop control effectively prevent significant overheat while reduces pumping power consumption. Among the proportional controllers, increasing the control parameter K_p reduces pumping power consumption while increasing the possibility of overheating. The controller of $K_p = 1.5$ obtains the best balance between the pumping power consumption and cooling performance.

Lastly, we train and test the proposed deep explicit MPC. In training stage, the neural network is built as a stack of six 64-width ReLU residual layers, one full-connected output layer and a λ -function layer implementing $\mathbf{h}_{1:n} \circ \mathbf{f}_{1:n}$, which has

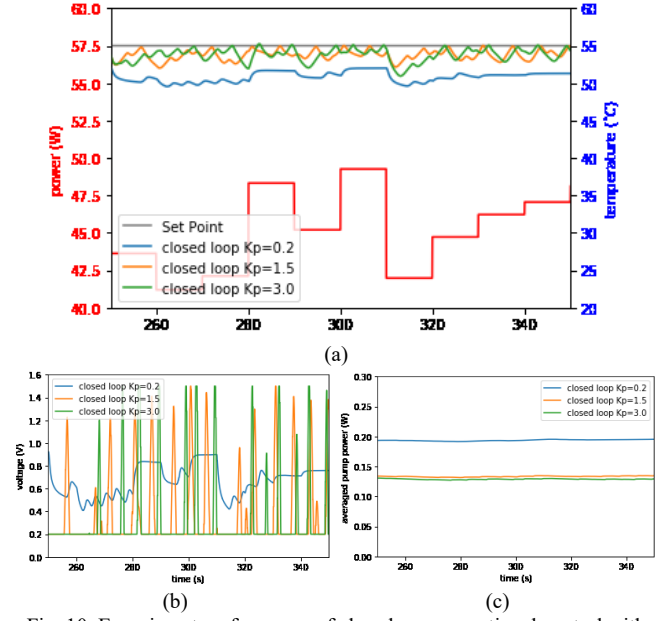


Fig. 10. Experiment performance of close loop proportional control with different parameters. (a) core junction temperature, (b) pump signal, (c) averaged pumping power.

24.4K parameters (the dimension of θ) in total. The data set of size $64K \times 55$ is fed in a batch of 64 for 32 epochs and training time within 150s on Colab is taken to train it to approximate a MPC with a horizon of $n = 25$. The model and weights after training is downloaded as a formatted file from the cloud to local server for testing.

The trained deep explicit MPC is tested in the same platform with same load schedule. The runtime computational cost is negligible for the local controller. Fig. 11 shows the comparison of the deep explicit MPC (DEMPC) with the open loop control and closed loop control. It can be found that the deep explicit MPC achieves more precise control of cooling while reducing the pumping power. The temperature is generally closer to the set point with minimal overheating while the pumping power consumption is further reduced.

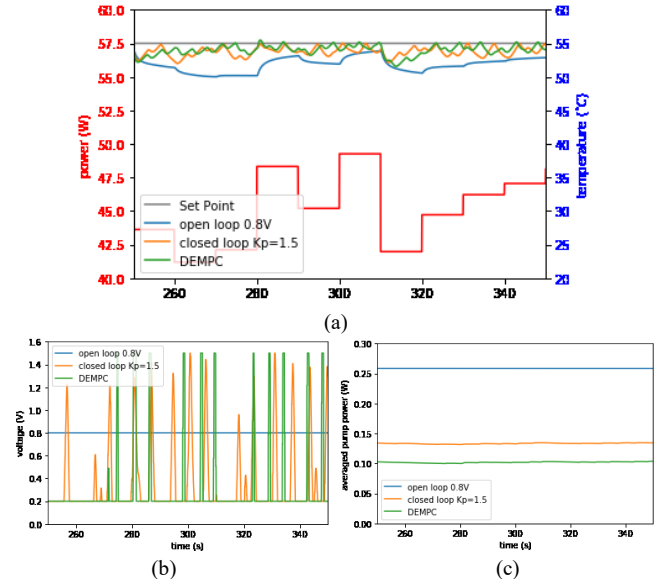


Fig. 11. Experiment performance of explicit model predictive control compared with open loop and closed loop control. (a) core junction temperature, (b) pump signal, (c) averaged pumping power.

Further quantification of the performance requires calculating the efficiency in terms of PUE and the range of overhear. The PUE is defined by:

$$\text{PUE} = \frac{\int_0^t P_{\text{pump}} dt}{\int_0^t P_{\text{load}} dt} + 1 \quad (19)$$

In this definition, the PUE only taken the pump in primary loop into consideration because peripheral energy consumptions in secondary loop, electric power delivery and etc. are out of the scope of this paper. This experiment reaches PUE 1.002346. These and the other parts of energy consumption are linearly addable to (19) if necessary, but this paper assumes fair background condition for all control strategies. The average overhear and the time percentage are defined by:

$$\text{Overheat} = \frac{1}{t} \int_0^t T - T_r d\tau \quad (20)$$

$$\text{Time\% Overheat} = \frac{1}{t} \int_0^t \delta(T - T_r > 0) d\tau \quad (21)$$

In the experiment, the deep explicit MPC achieves -0.962°C mean overhear degree level with 6.5% of time being overheated. Fig. 12 illustrates the PUE vs. Overheat curves for open loop control, closed loop control and deep explicit MPC. The vertical error bar shows the maximum and minimum overhear during the experiment. The circle at the center is colored by the time% overhear from blue to red. The open loop control with different pump voltage is shown in blue. The closed loop control with different K_p is shown in orange. It can be seen that the deep explicit MPC made almost same control performance with better efficiency. Comparing with open loop control under almost same energy budget (PUE), the cooling effectiveness of the deep explicit MPC successfully reduces the averaged degree of overhear by 1.21°C from 0.25°C to -0.96°C , and reduces over 8 times of the overhear time percentage from 57.4% down to 6.5%. And comparing with closed loop control under almost same average overhear, the deep explicit MPC saves energy consumption by 21.7% from 1.0031 to 1.0023.

VI. CONCLUSIONS

A design of control system for processor liquid cooling application is presented in this paper including architecture, modeling and control algorithm. The novel contribution of this work includes: 1) established two-stage control architecture to combine the strength of MATLAB, LabVIEW, Python and

Colab for data center liquid cooling application; 2) modeled processor liquid cooling package by the hybrid of non-linear steady-state response with linear transient state-space model to simplify computation; 3) approximated MPC explicitly by deep neural network for improving performance and reducing online computational cost; 4) validate the application of deep explicit MPC in the temperature control and disturbance rejection of data center liquid cooling system. The proposed deep explicit MPC achieves satisfactory cooling effectiveness and energy efficiency while satisfying limited computational power. Averaged PUE for the primary loop reaches 1.002346 with mean overhear degree -0.962°C . The deep explicit MPC improves the cooling effectiveness by reducing overhear level over 1.2°C and reducing overhear time percentage over 8 times of the traditional open loop control under similar energy budget, and reduces the energy consumption by 21.7% than close loop proportional control under similar average overhear level.

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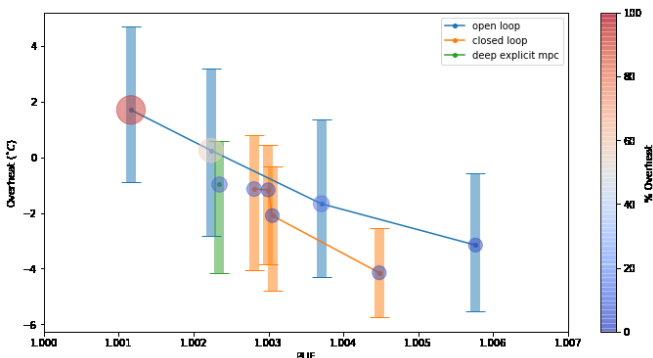


Fig. 12. PUE vs. Overheat characteristics of deep explicit MPC compared with open loop and closed loop controls.

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